

Project Design Phase-I
Proposed Solution

Date	26 February 2026
Team ID	699d5f82fdf409fdbd1bec66
Project Name	Malaria Detection using Deep Learning
Maximum Marks	2 Marks

Proposed Solution Template:

Project team shall fill the following information in proposed solution template.

S.No.	Parameter	Description
1.	Problem Statement (Problem to be solved)	Malaria is a life-threatening disease affecting millions of people worldwide. Traditional malaria diagnosis relies on manual microscopic examination of blood smear images by trained pathologists. This process is time-consuming, requires expert knowledge, and may lead to human errors. Therefore, there is a need for an automated system that can quickly and accurately detect malaria parasites in blood cell images.
2.	Idea / Solution description	The proposed solution is an AI-powered malaria detection system that uses deep learning algorithms to analyze microscopic blood smear images and determine whether the cell is infected or uninfected. The system works as follows: 1. Blood smear images are collected from the Kaggle malaria dataset. 2. Images are preprocessed and augmented to improve model training. 3. Multiple deep learning models are trained (Custom CNN, MobileNetV2 , EfficientNetB0) and the best performing model is selected. 4. A Flask web application allows users to upload blood smear images and receive instant predictions.
3.	Novelty / Uniqueness	• Uses deep learning for automated malaria detection. • Compares multiple CNN architectures to select the best model. • Provides real-time prediction via a web application. • Reduces dependency on manual microscopic examination. • Can be extended for other disease detection tasks.
4.	Social Impact / Customer Satisfaction	• Enabling early detection of malaria • Reducing diagnostic errors • Supporting doctors and laboratory technicians • Improving healthcare services in rural and remote areas • Saving lives through faster treatment decisions

5.	Business Model (Revenue Model)	Possible revenue streams include: • Licensing the software to hospitals and diagnostic laboratories • Providing AI diagnostic services through cloud platforms • Integrating the system into healthcare management software • Partnerships with medical device companies
6.	Scalability of the Solution	The system is highly scalable because: • It can process large volumes of medical images automatically. • The model can be deployed on cloud platforms. • It can be integrated with hospital information systems. • Future upgrades can support multi-disease detection.